

S001 - Seller API v2 Integration Project Plan

Project Document | Engineering

Document ID	PRJ-S001-002	Seller ID	S001
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2023-06-19
Related Scenario ID	Not scenario-specific	Document Type	Project Document

1. Project Overview

Field	Value
Project name	Aurelia Seller API v2 Migration
Project code	PRJ-S001-INT
Seller	S001 - Aurelia Home Decor
Objective	Replace CSV and manual processes with real-time API integration
Start date	2023-06-26
Target completion	2023-09-29
Engineering owner	Vikram Singh
Business sponsor	Aditya Verma
Seller counterpart	Ritika Bansal

This project removes the interim manual processes agreed during onboarding (decision OB-001) and moves Aurelia onto the real-time Seller API v2 integration before the festive peak.

2. Problem Statement

Aurelia went live in February 2023 on a CSV listing flow with a manual stock file uploaded twice daily. Six months of operation exposed three concrete problems.

- First, the stock file cadence means published availability can be up to twelve hours stale. During normal trading this is tolerable. During a promotional window it produces oversells, and Aurelia's long manufacturing lead times make an oversell expensive to unwind.
- Second, order notification by portal download and email requires a human in the loop at the seller's end. Dispatch SLA performance degrades on weekends and public holidays for no technical reason.
- Third, settlement statements retrieved manually cannot be reconciled automatically inside Aurelia's finance system, so every cycle requires manual matching. Aurelia's finance team has asked repeatedly for a machine-readable settlement feed.

3. Objectives and Success Criteria

Objective	Success criterion
Real-time inventory	Stock published within 60 seconds of a warehouse movement
Automated order intake	100 percent of orders delivered by webhook, polling only as fallback

Objective	Success criterion
Machine-readable settlement	Settlement report retrieved via API for every cycle
Reduced manual effort	Elimination of the twice-daily stock file
No regression	Dispatch SLA maintained at or above 96 percent through cutover

4. Scope

4.1 In Scope

- Seller API v2 credential provisioning and sandbox access for Aurelia.
- Inventory update endpoint integration from Aurelia's ERP.
- Webhook order push with signed payloads and a 15-minute polling fallback.
- Settlement report endpoint consumption.
- Catalogue maintenance through the API, with images continuing through the seller portal.
- Parallel-run and cutover planning.

4.2 Out of Scope

- Image pipeline migration; images remain a portal function.
- Returns automation; Aurelia continues to process returns manually.
- Second fulfilment node; no change to the single-node arrangement.
- Pricing automation; prices remain seller-managed through the portal.

5. Delivery Phases

Phase	Dates	Deliverable	Exit criterion
1. Enablement	26 Jun - 14 Jul	Sandbox credentials, API walkthrough	Seller can authenticate and call read endpoints
2. Inventory	17 Jul - 11 Aug	ERP to stock endpoint integration	72-hour sandbox run with zero drift
3. Orders	14 Aug - 01 Sep	Webhook receiver and polling fallback	Signed payloads verified, replay tested
4. Settlement	04 Sep - 15 Sep	Report endpoint consumption	Two historical cycles reconciled
5. Parallel run	18 Sep - 26 Sep	Both paths active	Variance below 0.5 percent for 7 days
6. Cutover	29 Sep	Manual processes disabled	Manual stock file retired

6. Technical Design Summary

6.1 Inventory

Aurelia's ERP emits a stock movement event on every pick, receipt and adjustment. A lightweight adapter translates these into Seller API v2 stock update calls. The adapter batches movements within a 30-second window to avoid excessive call volume on high-velocity SKUs, which keeps publication well inside the 60-second target.

A reconciliation job runs nightly and compares Aurelia's authoritative stock position against published availability. Any SKU with a variance is logged and re-pushed. This job is the control that makes the real-time path trustworthy.

6.2 Orders

Orders are delivered by webhook with an HMAC signature. The receiver acknowledges within two seconds and processes asynchronously. Because webhook delivery is at-least-once, the receiver deduplicates on order identifier before creating an ERP order.

The deduplication requirement was specified explicitly here because at-least-once delivery is a platform-wide characteristic. The same class of requirement is revisited in later platform work on refund processing.

A polling fallback runs every 15 minutes and fetches any order not seen by the webhook path in the previous 30 minutes.

6.3 Settlement

Aurelia retrieves a settlement report per cycle through the report endpoint. The report includes gross value, commission, reversals, adjustments and net payable, at order-line granularity. Aurelia's finance system matches this against its own order export and flags any line with a variance above one percent.

7. Risks

Risk	Likelihood	Impact	Mitigation
Cutover lands too close to festive peak	Medium	High	Cutover fixed at 29 September, freeze after 10 October
ERP adapter throughput insufficient	Low	Medium	30-second batching window, load tested at 3x peak
Webhook endpoint availability at seller	Medium	Medium	Polling fallback retained permanently
Settlement report mismatch	Medium	High	Two historical cycles reconciled before cutover
Seller-side resource contention	Medium	Medium	Named counterpart, weekly checkpoint

8. Governance

A weekly checkpoint is held each Tuesday with Vikram Singh, Aditya Verma and Ritika Bansal. Phase exit requires written confirmation from both the engineering owner and the seller counterpart. Any slip to the cutover date must be escalated to Business, because the festive freeze makes late delivery materially more expensive than a scope reduction.

9. Outcome

The project completed on 30 September 2023, one day after target. All phase exit criteria were met. The manual stock file was retired, dispatch SLA improved from 96.4 percent to 98.1 percent in the following quarter, and settlement reconciliation moved from manual matching to automated variance detection at Aurelia's end.

The nightly inventory reconciliation job introduced in phase 2 has since run without a material finding, which is one reason inventory has never been a significant issue on this account.